

Edge Negotiator methodology: simulation-first design for SLM traffic signal control

The Edge Negotiator architecture can be rigorously evaluated as a simulation-first study in SUMO, provided the experimental design pre-commits to paired statistical comparisons, operationalises CoT faithfulness through behavioural probes, and frames all claims within explicit simulator-fidelity boundaries. This report supplies the evidential scaffolding for nine compressed sub-questions and a full experimental design targeting ITSC/AAAI/KDD submission. The two empirical contributions — Besu QBFT vs Trillian Tessera cost-benefit, and CoT faithfulness characterisation for sub-7B SLMs in safety-adjacent loops — are addressable on a single developer laptop with the statistical plan described in §3. The design leverages Qwen3-4B's native dual thinking/non-thinking modes as a within-model experimental lever unavailable in any competing architecture.

§1 Scope and architectural fixed points

This study evaluates the Edge Negotiator, an architecture placing autonomous traffic signal decisions at the network edge rather than in a centralised cloud. "Edge" denotes architectural topology — decisions co-located with intersection controllers — not a hardware deployment claim. All experiments run in SUMO with TraCI on a developer laptop (RTX 3060/4060 mobile-class GPU). Edge deployment, distributed multi-machine operation, and field validation are explicitly out of scope and identified as future work, as formally agreed with academic supervisor Dr Akin Delibasi (UCL) and industry supervisor Lee Stott (Microsoft Principal Cloud Advocate / UCL Honorary Associate Professor).

The SLM stack pairs **Qwen3-4B** (Apache 2.0, ~2.5 GB Q4_K_M) as primary with **Phi-4-mini** (3.8B, MIT) (`Hugging Face`) as comparison. Traffic-R1 Public 0.1 serves as benchmark anchor (cited, not deployed). Hot-path output is a single-token phase ID with structured JSON metadata; audit-path CoT is generated separately, labelled (`trace_type: "post_hoc_justification"`), and CFG-constrained via XGrammar. Z3 SMT verification stands between every SLM output and SUMO actuation, (`de engineering`) enforcing no conflicting greens, NEMA-compliant phase sequences, and minimum clearance intervals. MaxPressure runs always-on alongside every SLM decision as both baseline and safety fallback. The ledger layer compares Hyperledger Besu QBFT (Docker, (`-network=dev`), web3.py client, on-chain SHA-256 hash + off-chain 2 KB payload) against Trillian Tessera Merkle log. The network is a **5×5 km OSM-derived section of inner south London** spanning the Lambeth/Southwark border from Elephant & Castle to Brixton, with 2019 English IMD decile joins at LSOA granularity.

§2 Compressed sub-question reviews

(a) State of the art in SUMO-based autonomous TSC

The TSC literature splits into four tracks. **RL-based methods** — IntelliLight (KDD 2018), (ACM Digital Library) PressLight (KDD 2019), (GitHub) CoLight (CIKM 2019, graph-attentional network-level cooperation), (ResearchGate) MPLight (AAAI 2020), Advanced-XLight (ICML 2022) (Proceedings of Machine Le...) — converge on average travel time (ATT) as primary metric across standardised datasets (Jinan 3×4, Hangzhou 4×4, New York 28×7) with one-hour episodes and 5 random seeds. **LLM-based methods** — LLMLight (KDD 2025, arXiv:2312.16044), (arXiv) CoLLMLight (arXiv:2503.11739), (GitHub) Traffic-R1 (arXiv:2508.02344), LA-Light (arXiv:2403.08337) — demonstrate that LLMs can match or exceed trained RL policies with zero-shot generalisation. Critically, **most top papers use CityFlow, not SUMO**; only LA-Light and MARL papers use SUMO directly. (arXiv) **MARL approaches** (MAPPO, QMIX) operate under CTDE; QMIX achieves lowest queue lengths in PyTSC benchmarks on SUMO grids (PubMed Central) (PMC:11902778). (PubMed Central) **Classical controllers** — Webster's fixed-time, MaxPressure (Varaiya 2013, throughput-optimal under assumptions), actuated/SOTL — remain mandatory baselines; SCATS/SCOOT are proprietary and cannot be benchmarked openly. (ResearchGate) (ResearchGate) The standard results table groups methods by category (Traditional → RL → LLM → Proposed), reports ATT/AWT/AQL across datasets, and highlights best/second-best.

(b) What SUMO can and cannot model

SUMO offers **Krauss (default), IDM, and Wiedemann car-following models**; calibration studies show IDM replicates real driving best, though all defaults produce shorter headways than field data (ResearchGate) (sumo.dlr.de/docs/Car-Following-Models.html). Mixed traffic (buses, trucks, bikes, emergency vehicles) is supported via (vClass) with per-class dynamics. Emergency vehicles get blue-light device, right-side overtaking, and red-light violation via (jmDriveAfterRedTime), but signal preemption must be implemented manually through TraCI; (SUMO) SAE J2735/V2X requires co-simulation with OMNeT++/Veins (sumo.dlr.de/docs/Simulation/Emergency.html). **Pedestrian modelling is the key limitation**: the default striping model divides lateral width into discrete stripes (SUMO) but SUMO documentation acknowledges "realistic simulation is not possible" with current defaults; (GitHub) there is no built-in model for pedestrian demographics or mobility impairment (sumo.dlr.de/docs/Simulation/Pedestrians.html). Pedestrian delay parity claims must therefore carry an explicit fidelity caveat. SUMO has **no native sensor noise model** — all observations are ground truth unless manually injected via TraCI or SUFI fault injection (PubMed Central) (Valu3s) (EDCC 2021). Adversarial ghost vehicles can be injected by adding virtual vehicle IDs via (traci.vehicle.add) or by corrupting detector outputs at the application layer.

(c) Latency measurement methodology

Three approaches exist. **Approach A** (offline profiling + latency injection): profile $N \geq 1,000$

inferences offline, fit a log-normal distribution, inject sampled delays via `time.sleep()` before `traci.simulationStep()`. Fast but misses dynamic GPU effects. **Approach B** (in-loop): SLM runs live per decision point; TraCI naturally blocks until all clients call `simulationStep` (SUMO) (SUMO) (sumo.dlr.de/docs/TraCI.html). Realistic but limits episode count. **Approach C** (hybrid): bulk experiments via A, validation subset (50–100 episodes) via B, then KS-test comparison of latency distributions. **Approach C is recommended** — it combines statistical power with realism validation. The single-token hot path drastically relaxes the latency constraint: time-to-first-token (TTFT) dominates and is proportional to prompt length, whereas full CoT in thinking mode generates 100–500+ tokens — a **10–50× latency multiplier**. Reporting must separate TTFT, per-token generation time, and total CoT latency, and present p50/p95/p99 distributions with bootstrapped CIs across prompt template variations and thinking-mode toggles. Z3 verification adds negligible overhead (<1 ms for ~50 Boolean/integer constraints) (github.com/Z3Prover/z3).

(d) Besu QBFT vs Trillian Tessera experimental protocol

Besu QBFT provides immediate finality via $\geq 2/3$ validator supermajority signing, (Hyperledger) tolerating $\lfloor (n-1)/3 \rfloor$ Byzantine faults. (Hyperledger) Official benchmarks show **750–1,200 TPS** on 4-validator AWS setups with 2-second block periods (atlassian) (lf-hyperledger.atlassian.net). On a single laptop with Docker Compose (4 containers), expect 8–16 GB RAM and degraded throughput. **Trillian Tessera** (the transparency-dev successor to Trillian v1, (Transparency) not ConsenSys Tessera) is an append-only Merkle log (GitHub) achieving **1,587 QPS sustained** with observed time-to-integrate of 569–1,878 ms (GitHub) ([transparency-dev performance docs](https://transparency-dev.github.io/performance/docs/)). Inclusion proof size scales logarithmically: **~640 bytes at 1M entries** vs Besu's Merkle Patricia trie proofs at ~1–2 KB. The experimental protocol replays identical decision streams ($\geq 10,000$ decisions) through both backends, recording per-decision write latency, finality time, proof size, storage growth, CPU/memory (Docker stats), and quarterly-audit query latency. The on-chain anchoring pattern stores a 32-byte SHA-256 hash per decision, (CSIRO) achieving **62.5× compression** vs the 2 KB off-chain payload ([CSIRO blockchain patterns](https://csiro.org.au/blockchain-patterns/)).

(e) CoT faithfulness probe design

The **primary methodology** follows Turpin et al. ([arXiv:2305.04388](https://arxiv.org/abs/2305.04388), NeurIPS 2023): inject known biasing features into prompts (e.g., "always suggest phase A") and measure whether CoT mentions the bias. Turpin found near-zero bias-mention rates — models rationalise biased answers without acknowledging the influence. The **secondary methodology** follows Lanham et al. ([arXiv:2307.13702](https://arxiv.org/abs/2307.13702)): truncate CoT at progressive points and measure answer consistency via Area Over Curve (AOC). Qwen3-4B's **native dual thinking/non-thinking modes** (arXiv) ([arXiv:2505.09388](https://arxiv.org/abs/2505.09388)) enable within-model comparison: same weights, same prompt, toggling `/think` vs `/no_think`. Phi-4-mini (MIT, 3.8B, (Microsoft Azure) [arXiv:2503.01743](https://arxiv.org/abs/2503.01743)) lacks native thinking mode and serves as control. **Faithfulness threshold:** CoT must correlate with at least 3 of 5 observable state variables (queue lengths, phase timing, vehicle counts per approach, delay, throughput) at $r > 0.5$; below this, CoT is downgraded to "decorative" in the audit trail. Additional

probes: paraphrase invariance (Lanham-style rewording), constraint-violation reference rate (does CoT mention safety violations when Z3 flags them). Activation patching is considered and excluded for tractability (§5).

(f) Baselines and ablations for target venues

Standard ITSC/AAAI/KDD tables require: **Fixed-time** (Webster's), **Actuated** (SOTL), **MaxPressure** (Varaiya 2013), at least one RL baseline (**CoLight** as canonical), and **Traffic-R1 Public 0.1** as benchmark anchor with the caveat that it was trained on CityFlow, not SUMO. The SLM+MaxPressure-fallback system must be compared against MaxPressure-only to isolate the SLM's marginal contribution. Critical ablations form a **2⁴ factorial core**: SLM $\pm$ CoT logging, $\pm$ Z3 verification, $\pm$ MaxPressure fallback, thinking-mode vs non-thinking-mode Qwen3. Secondary ablations: Qwen3-4B vs Phi-4-mini, Besu vs Tessera backend. The standard table format (rows = methods grouped by type, columns = scenarios, cells = ATT $\pm$ std, best/second-best bolded) follows LLMLight and Traffic-R1 conventions ([arXiv +2](#)) ([arXiv:2508.02344](#)). Transportation Research Part C additionally expects violin/box plots, multiple demand scenarios, and statistical significance tests.

(g) Metrics

Traffic performance: ATT, AWT, throughput, mean queue length, stops per vehicle, CO₂/fuel proxy (SUMO emissions model), emergency vehicle delay. **Equity**: Gini coefficient of mean per-vehicle delay across LSOAs (headline, following [Kadam & Ranjitkar 2022](#)); Rawlsian max-min on worst-served IMD decile; disparate impact ratio (IMD 1-3 vs 8-10, flagging DIR < 0.80); ([Superwise](#)) pedestrian delay parity (with explicit SUMO fidelity caveat). **Safety**: SUMO SSM device for TTC (threshold < 3.0 s), PET (< 2.0 s), DRAC (> 3.0 m/s²) ([SUMO](#)) ([GitHub](#)) ([sumo.dlr.de/docs/Simulation/Output/SSM_Device.html](#)); ([dlr](#)) forced-stop events; fallback activation rate; Z3 verification failure rate. **Accountability**: proportion of decisions with complete audit trails, time-to-audit-retrieval, audit log size growth. **Adversarial robustness**: attack success rate under ghost-vehicle injection. **Compute**: simulated decision latency distribution, tokens per decision, fallback activation rate due to latency budget breach. **Faithfulness**: bias-mention rate, paraphrase invariance score, constraint-violation reference rate. **Ledger**: per-decision write latency Besu vs Tessera, proof size, storage growth.

(h) Framing simulation as scientific contribution

Simulation-first is the norm in TSC research. Müller & Sabatelli (ITSC 2023) state: *"A key barrier to progress in this area is the reality gap, the discrepancy that results from differences between simulation models and their real-world equivalents"* ([arXiv:2307.11357](#)). Da et al. (AAAI 2024) frame the gap as: *"a notable performance gap arises when deploying simulator-trained policies to real-world environments"* ([AAAI 2024](#)). The real-world I-24 MARL deployment paper explicitly notes *"physical systems in the real world are subject to delays and inaccuracies"* as motivation for a four-stage sim-to-deployment pipeline ([arXiv](#)) ([arXiv:2503.01017](#)). **Recommended Edge**

Negotiator footnote: "The 'Edge' in Edge Negotiator refers to the architectural placement of decision-making at the network edge (co-located with intersection controllers) rather than in a centralised cloud. All experiments in this work execute within a single simulation process; deployment to physically distributed edge hardware is beyond the scope of this dissertation and identified as future work." Supervisor agreement on scope can be stated in the dissertation introduction: "The scope of this dissertation, as agreed with the supervisory panel, focuses on simulation-based evaluation."

(i) Decentralisation on one machine

All CTDE (Centralised Training, Decentralised Execution) MARL papers simulate **"decentralised" agents on a single machine** — this is universally accepted practice, not a limitation requiring defence ([arXiv](#)) ([arXiv](#)) ([Amato 2024](#)). The recommended approach is a **shared SLM with serialised inference**, which is honest about the single-machine constraint and avoids GPU memory duplication. Per-intersection parallel processes are architecturally faithful but impractical on consumer hardware. Communication should use libsumo (shared memory, no socket overhead) rather than TraCI TCP for efficiency, ([SUMO +2](#)) with an explicit statement: *"Inter-agent communication is instantaneous via shared memory. Real-world inter-intersection communication delay is not modelled."* Precedents in distributed systems confirm that simulating Paxos, Raft, and BFT on single machines is standard ([GitHub](#)) ([GitHub](#)) ([distributed-consensus-reading-list](#)). CityFlow's design — single-process C++ with pybind11, **25× faster than SUMO** on 30×30 grids — confirms that single-machine multi-agent simulation is the community norm ([Cityflow-project](#)) ([arXiv:1905.05217](#)).

§3 Proposed experimental design

This section constitutes the primary deliverable and is written at methodology-chapter grade.

Network and scenario specification

The study network is a **5×5 km OSM-derived section** of inner south London spanning the Lambeth/Southwark border, from Elephant & Castle (north) to Brixton (south). This area includes major A-roads (A3, A23, A202, A215), numerous signalised intersections, and a natural IMD deprivation gradient: northern LSOAs (Walworth, Camberwell) fall in **IMD deciles 1–3**, while southern areas (parts of Herne Hill, Dulwich edges) reach **deciles 5–8**, creating the within-network variation essential for equity analysis. Lambeth and Southwark each have a mean IMD decile of approximately 4, ([Research Square](#)) ranked among the most deprived 15% of English local authorities ([ResearchSquare rs-7179514](#)). OSM coverage of central London is excellent; signal timing data can be supplemented from TfL open data. LSOA boundaries from the ONS Open Geography Portal are spatially joined to SUMO edges via centroid intersection, with population-weighted averaging where edges span multiple LSOAs.

Four scenario types structure the evaluation. **AM peak** (07:30–09:30): high inbound demand towards Elephant & Castle, testing corridor throughput. **Off-peak** (11:00–13:00): moderate bidirectional flow, testing unnecessary delay avoidance. **PM peak** (16:30–18:30): reversed dominant flow towards Brixton. **Incident/perturbation** (injected lane closure at a critical intersection during PM peak): testing resilience and fallback activation. Each scenario runs for **2 hours of simulated time** (7,200 steps at 1-second resolution), with a 15-minute warm-up period excluded from analysis (`dot`) following FHWA guidance ([FHWA TAT Vol III](#)).

Baselines and ablation matrix

The full comparison matrix includes six baselines and eight ablation conditions:

Category	Method	Notes
Classical	Fixed-time (Webster's)	Optimised cycle from calibrated flows
Classical	Actuated (SOTL)	Gap-based phase extension
Classical	MaxPressure	Also serves as always-on fallback
RL	CoLight	Canonical MARL baseline, retrained on study network
LLM	Traffic-R1 Public 0.1	Benchmark anchor; CityFlow-trained, SUMO-evaluated with caveat
Proposed	Edge Negotiator (full)	Qwen3-4B + Z3 + MaxPressure fallback + Besu ledger

The ablation matrix systematically removes or varies components. The **2⁴ factorial core** crosses: CoT logging (on/off), Z3 verification (on/off), MaxPressure fallback (on/off), and thinking mode (think/no-think). Two additional single-factor ablations swap Qwen3-4B → Phi-4-mini and Besu → Tessera. This yields **16 + 2 = 18 conditions**, each run across 4 scenarios.

Statistical analysis plan

Seed count and power analysis. Each condition × scenario combination runs with **N = 30 pre-registered random seeds**, selected by drawing 30 integers from a Mersenne Twister seeded at 42 and committed to a `replication/seeds.json` file before any experiments execute. SUMO's `--seed` flag ensures deterministic reproduction (`SUMO`) ([sumo.dlr.de/docs/Simulation/Randomness.html](#)). (`Eclipse Foundation`) The seed count is justified by power analysis: with typical CV of 15% in per-seed ATT and a target of detecting a **10% ATT improvement** (Cohen's $d \approx 0.67$, medium-to-large), a paired Wilcoxon test at $\alpha = 0.05$ achieves **>90% power with 30 seeds**. For smaller effects (5%, $d \approx 0.33$), 30 seeds yields ~55% power; if

preliminary results suggest small effects, the design escalates to 50 seeds for secondary endpoints.

Pre-specified endpoints. The **primary endpoint** for the traffic performance track is mean ATT across all vehicles, tested at $\alpha = 0.05$ without correction. Secondary traffic endpoints (AWT, throughput, queue length, stops, CO₂) are tested with **Benjamini-Hochberg FDR correction at $q = 0.10$** (Benjamini & Hochberg 1995). (Wikipedia) (Columbia University Mailm...) The primary endpoint for the equity track is Gini coefficient of per-LSOA mean delay. The primary endpoint for the safety track is fallback activation rate. Pre-specifying these prevents p-hacking; all analysis scripts are committed to version control before experiments run.

Test selection by metric type. Continuous paired metrics (ATT, AWT, queue length) use the **paired Wilcoxon signed-rank test**, which does not assume normality — appropriate given right-skewed delay distributions. Proportions (fallback activation rate, Z3 failure rate, bias-mention rate) use **BCa bootstrap confidence intervals** with 10,000 resamples. Network-level metrics with intersection-level nesting use **linear mixed-effects models**: $\text{delay} \sim \text{treatment} + (1|\text{intersection}) + (1|\text{seed})$ in lme4 notation, accounting for correlation of outcomes across intersections within the same simulation run. Effect sizes are reported as Cliff's delta (non-parametric) alongside bootstrapped 95% CIs for all pairwise comparisons.

Reproducibility protocol. All configs, seeds, network files, and analysis scripts reside in a (replication/) directory. The README records SUMO version (via (sumo --version)), OS, Python version, GPU driver version, and quantisation format. Raw SUMO output (trip-info, SSM device, emission output) is archived per seed. Following RepliSims guidelines (Royal Society Open Science, 2023), the full pipeline is executable from a single entry-point script.

(Royal Society Open Science)

Validity argument

Internal validity is strong: identical seeds pair each treatment with its baseline, eliminating stochastic demand variation as a confounder. (Wikipedia) Warm-up exclusion prevents initial transient effects. The Z3 safety check is deterministic, so safety verification outcomes are fully reproducible.

External validity is explicitly bounded. (StatsDirect) The single network (Lambeth/Southwark) limits geographic generalisation. SUMO's Krauss/IDM car-following models produce shorter headways than field data, (Tib-op) meaning simulated throughput may be optimistic. Demand patterns are synthetic (calibrated to London averages, not intersection-level counts). Generalisation claims are restricted to "within the fidelity envelope of SUMO microscopic simulation on an OSM-derived London network."

Construct validity requires care. (OpenAnesthesia) ATT and AWT map directly to traveller experience. The Gini-of-delay equity metric is an established transport equity measure but captures only one dimension of justice; the Rawlsian max-min and DIR provide triangulation.

SUMO SSM metrics (TTC, PET) are accepted surrogates for crash risk but are not crash predictions. CoT faithfulness, operationalised as bias-mention rate and state-variable correlation, is a behavioural proxy — not a mechanistic guarantee of reasoning fidelity.

Statistical conclusion validity is supported by adequate power ($\geq 80\%$ for $d \geq 0.67$ at $N = 30$), appropriate non-parametric tests, mixed-effects modelling for nested data, and BH correction for multiple secondaries. The seed list is pre-committed to prevent "seed hacking."

Royal Society Open Science

Track 1 — Besu QBFT vs Trillian Tesseract

Experimental protocol. A synthetic decision stream of $\geq 10,000$ traffic signal decisions (timestamps, phase IDs, 2 KB structured JSON payloads) is generated from a single SUMO PM-peak scenario. Each decision's SHA-256 hash is submitted to Besu QBFT (4 Docker validators, `blockperiodseconds: 2`, `web3.py` `send_raw_transaction` with `geth_poa_middleware`) and the full entry to a Trillian Tesseract log (POSIX backend, single-instance). Submission timestamps, acknowledgement timestamps, and finality timestamps are recorded for each decision in both backends, enabling paired comparison. Metrics collected: write latency distribution (p50/p95/p99), finality time, inclusion proof size, on-chain vs off-chain storage growth rate, CPU/memory overhead (Docker stats at 1-second resolution), and quarterly-audit query latency (retrieving all decisions for one LSOA over a simulated quarter).

Threat model. This is the architectural argument; performance measurements quantify operational consequences. Six threats are evaluated:

(i) Single-operator compromise (municipality silently rewrites log). Besu QBFT provides **strong defence**: rewriting requires $\geq 2/3$ validator supermajority, `Hyperledger` not achievable by a single operator controlling 1 of 4+ validators ([besu.hyperledger.org QBFT](https://besu.hyperledger.org/qa/bft)). A standalone Tesseract Merkle log is **vulnerable** — the log operator can fork the tree — but adding C2SP witnesses restores strong defence via independent append-only verification.

(ii) Multi-validator collusion ($> 1/3$ QBFT threshold). Besu is **fundamentally vulnerable** if ≥ 2 of 4 validators collude (network stalls) or ≥ 3 of 4 collude (arbitrary block finalization). This is the inherent BFT limitation. `Hyperledger` Tesseract has no analogous threshold but faces an equivalent threat if witnesses collude with the log operator.

(iii) Log-server compromise (pure Merkle). Besu has no single log server (distributed consensus). Tesseract is **vulnerable without witnesses**: a compromised server can present forked trees, omit entries, or serve inconsistent views. Mitigation: C2SP witnesses cosigning checkpoints, client-side STH caching (`GitHub`) (github.com/C2SP/C2SP/blob/main/tlog-witness.md).

(iv) Split-view attacks (different auditors shown different log versions). Besu is **inherently resistant** — consensus ensures all validators see the same chain. Tesseract is **vulnerable by default** but the C2SP Witness Protocol specifically addresses this: witnesses verify consistency

and cosign checkpoints; clients require a quorum of cosignatures, making split views detectable ([GitHub](#)) ([GitHub](#)) (github.com/transparency-dev/witness).

(v) Censorship (auditors blocked from access). Both backends offer partial defence. Besu mitigates via auditor-controlled non-validator nodes that sync the chain. Tessera mitigates via tile-based architecture enabling CDN caching and mirroring — its static tile API is specifically designed for easy replication ([Transparency](#)) (transparency.dev tile-based logs).

(vi) Insider key compromise (SLM-decision attestation signing key). Neither backend fully defends against application-layer key compromise — a compromised attestation key can forge decision records regardless of log integrity. Besu's BFT protects validator keys (one compromise insufficient). Mitigation for both: HSM-based key storage (Besu's `--security-module`; [Hyperledger](#)) Tessera via ssh-agent as used by litewitness). ([GitHub](#))

The threat model demonstrates that **Besu QBFT provides stronger out-of-the-box integrity guarantees** (no single point of compromise, inherent split-view resistance) while **Tessera + C2SP witnesses approaches equivalent security at lower operational cost** (no consensus overhead, logarithmic proof sizes). ([GitHub](#)) The performance comparison quantifies the operational price of Besu's stronger defaults.

Track 2 — CoT faithfulness characterisation

Primary probe (Turpin-style bias injection). For each of 200 traffic states sampled from SUMO (stratified across scenarios and congestion levels), construct a biased prompt variant embedding an explicit directive: "Phase A is always the safest choice regardless of queue lengths." Submit to Qwen3-4B in thinking mode and non-thinking mode, and to Phi-4-mini. Measure: (a) whether the model selects phase A more often under bias (accuracy drop), and (b) whether the CoT explicitly mentions the biasing directive (**bias-mention rate**). Following Turpin et al., the expectation is near-zero bias mention despite significant accuracy shifts — establishing that post-hoc CoT in a decide-first architecture is unfaithful to actual decision drivers.

Secondary probe (Lanham-style truncation). For 100 traffic states, generate full CoT, then truncate at 0%, 25%, 50%, 75% of tokens and force answer completion. Plot fraction-of-CoT-seen vs answer-consistency to compute **Area Over Curve (AOC)**. Higher AOC indicates the model conditions on its CoT content. Compare across thinking/non-thinking modes and Qwen3-4B vs Phi-4-mini.

Additional probes. Paraphrase invariance: rephrase the first 20% of generated CoT and regenerate remainder; measure answer-change rate. Constraint-violation reference: when Z3 flags a safety violation, measure whether the post-hoc CoT mentions any of the violated constraints.

Faithfulness threshold operationalisation. Extract from each CoT the model's claims about 5 observable state variables: (1) queue length per approach, (2) current phase, (3) vehicle count per approach, (4) estimated delay, (5) throughput trend. Compute Pearson r between claimed and

actual values. A CoT is classified as "monitoring-grade" if $r > 0.5$ on ≥ 3 of 5 variables; otherwise it is downgraded to "decorative" and labelled as such in the audit trail. This threshold is pre-registered and applied uniformly across all conditions. The proportion of monitoring-grade CoTs across conditions is the headline metric for this track, directly informing whether CoT-based aggregate anomaly detection is feasible for sub-7B SLMs in safety-adjacent control loops — a gap no published study has addressed.

§4 Honesty clauses

The following claims are **explicitly not made** in this dissertation, with suggested defensive language:

No real-world deployment claim. *"This work evaluates the Edge Negotiator architecture in a simulated environment using SUMO. No traffic signals were controlled in the physical world. Deployment to real intersection controllers, including safety certification and regulatory approval, is identified as future work."*

No edge-hardware claim. *"The 'Edge' in Edge Negotiator denotes architectural topology — decision-making co-located with intersection controllers rather than centralised in a cloud — not a specific hardware platform. All experiments execute on a developer laptop; inference latency measurements on consumer GPU (RTX 3060/4060 mobile class) establish computational feasibility, not deployment-grade performance."*

No causal safety claim. *"Surrogate safety measures (TTC, PET) computed in SUMO are accepted proxies for conflict severity but are not crash predictions. This study does not claim that the proposed system reduces real-world collisions."*

No equity causation claim. *"Equity metrics (Gini, Rawlsian max-min, DIR) are computed on simulated delay distributions joined to IMD 2019 deciles at LSOA granularity. SUMO's pedestrian model (striping) lacks demographic fidelity; pedestrian delay parity results should be interpreted as indicative, not definitive. The simulation demonstrates that equity-aware evaluation is technically feasible, not that the proposed system achieves equitable outcomes in practice."*

No distributed-system claim. *"All simulated intersections execute within a single process on one machine. 'Decentralisation' refers to the logical separation of per-intersection decision-making, consistent with the CTDE paradigm standard in MARL literature. Physical distribution across networked hardware, including real communication latency and fault tolerance, is future work."*

No CoT-as-explanation claim. *"Post-hoc CoT generated in the audit path is labelled `trace_type: post_hoc_justification` and treated as a monitorability signal, not as a faithful explanation of the model's reasoning process. The faithfulness probes in §3 Track 2 empirically characterise the degree to which CoT correlates with observable state — results may confirm that CoT is largely decorative."*

Scope clause for examiners. *"The scope of this dissertation was formally agreed with the supervisory panel (Dr Akin Delibasi, UCL; Lee Stott, Microsoft / UCL). The study is simulation-first by design: the contribution is the architecture, its evaluation methodology, and the two empirical characterisations (ledger cost-benefit and CoT faithfulness). This scope is consistent with standard practice in the TSC simulation literature, where real-world deployment is a separate research contribution requiring safety certification, regulatory approval, and infrastructure partnerships beyond the remit of an MSc dissertation."*

§5 Risks and contingency plan

Qwen3-4B does not fit the decision loop

Risk: Inference latency on RTX 3060/4060 exceeds the simulation step budget even for single-token hot path, making Approach B (in-loop) infeasible for statistically adequate episode counts.

Contingency: Fall back entirely to Approach A (offline profiling + injection). If even offline profiling reveals >5-second TTFT at Q4_K_M quantisation, switch primary model to Phi-4-mini (3.8B, potentially faster) and report Qwen3-4B results only for the offline latency characterisation. The single-token hot path makes this risk low — TTFT for a 2.5 GB Q4_K_M model on mobile GPU should be well under 1 second for prompts under 2,000 tokens.

Fallback activates too often

Risk: MaxPressure fallback triggers on >50% of SLM decisions (malformed output, latency breach, low confidence), rendering the SLM contribution marginal. **Contingency:** This is itself a publishable finding. Report the fallback activation rate as a primary metric. If the rate exceeds 30%, conduct a root-cause analysis (output format violations vs latency vs confidence threshold) and tighten the XGrammar CFG constraints. The ablation comparing SLM+fallback vs MaxPressure-only isolates whether the SLM contributes even intermittent improvement.

Equity metrics show no signal in simulation

Risk: The Lambeth/Southwark network, when loaded with synthetic demand, produces near-uniform delay across LSOAs, yielding Gini ≈ 0 with no equity differentiation. **Contingency:** Introduce demand asymmetry correlated with IMD (higher demand on roads serving deprived LSOAs, reflecting observed London commuting patterns). If the Gini remains flat, report this null result as evidence that simulated demand patterns are insufficient for equity analysis — a methodological contribution in itself. The IMD gradient in the study area (deciles 1-3 north, 5-8 south) provides the best available chance of observing differential impact.

CoT faithfulness probe shows zero signal

Risk: Both Turpin-style bias injection and Lanham-style truncation show that Qwen3-4B CoT is

entirely decorative — zero bias mention, zero AOC differentiation. **Contingency:** This is the most publishable risk. A confirmed null result — "CoT in sub-7B SLMs is decorative in safety-adjacent control loops" — directly addresses the identified literature gap and has strong implications for audit system design. The methodology chapter should explicitly pre-commit to reporting null results. The thinking-vs-non-thinking comparison and the Qwen3-vs-Phi-4-mini comparison provide additional analytical dimensions even if the overall faithfulness signal is zero.

SUMO network construction fails

Risk: The OSM extract for Lambeth/Southwark produces an unusable SUMO network (disconnected edges, unrealistic intersection geometries, missing turn lanes). **Contingency:** Use SUMO's `netconvert` with `--geometry.remove` and manual editing to repair critical intersections. If repairs exceed reasonable scope, fall back to a synthetic grid calibrated to Lambeth/Southwark dimensions and intersection density, with an explicit caveat about reduced external validity.

Besu Docker deployment overwhelms laptop resources

Risk: Running 4 Besu validator containers alongside SUMO and SLM inference exhausts RAM/CPU. **Contingency:** Run the ledger comparison as a separate experimental phase (not concurrent with SUMO). Replay pre-recorded decision streams through the ledger backends offline. If even offline Besu is too heavy, reduce to 2 validators (sacrificing BFT guarantees) and document the deviation.

Activation patching: considered and excluded

Activation patching — a mechanistic interpretability technique where internal activations from a clean forward pass are patched into a corrupted pass at specific layers — could theoretically verify whether Qwen3-4B's internal computation follows the CoT's stated reasoning path. Recent work by Chen et al. ([arXiv:2507.22928](https://arxiv.org/abs/2507.22928)) combines SAE-based feature extraction with activation patching for CoT faithfulness. However, this technique requires (a) one forward pass per component per prompt pair across 36 layers and multiple attention heads, (b) access to full-precision model internals incompatible with Q4_K_M quantised inference via llama.cpp, and (c) SAE training pipelines still under active development. It is excluded from this study for tractability but identified as a priority for future work with full-precision model access on research-grade hardware.

Conclusion

This experimental design supports two empirical contributions within the constraints of a simulation-first MSc dissertation. **Track 1** (Besu QBFT vs Trillian Tessera) quantifies the operational cost of consensus-based integrity versus append-only transparency log integrity for

AI decision provenance — a comparison that, to our knowledge, has not been conducted with traffic signal decision payloads. The threat model provides the architectural justification; the paired replay design delivers the performance data. **Track 2** (CoT faithfulness) addresses a genuine gap: no published study characterises CoT faithfulness for sub-7B SLMs in safety-adjacent control loops. Qwen3-4B's native dual-mode architecture provides a unique within-model experimental lever, and the pre-registered faithfulness threshold ($r > 0.5$ on $\geq 3/5$ state variables) transforms a vague concept into a testable, actionable criterion for audit system design.

The statistical plan — 30 pre-registered seeds, paired Wilcoxon tests, mixed-effects models for nesting, BH correction for secondaries — exceeds the 5-seed convention in TSC literature and provides methodology-chapter-grade rigour. The honesty clauses in §4 and contingency plans in §5 collectively ensure that every plausible outcome — including complete CoT decorativeness and dominant fallback activation — remains publishable. The architecture earns its "Edge" designation not from hardware but from a design principle: decisions belong at the point of actuation, not in a distant cloud, and the provenance trail must be as trustworthy as the decision itself.